

## Independent events



- 1 Write an equation for  $P(A \text{ and } B)$  in terms of  $P(A)$  and  $P(B)$  given that the two events are independent.
- 2 A coin is tossed twice.
  - a Construct a tree diagram showing the results of the given experiment.
  - b Use the tree diagram to find the probability of getting:
    - i Exactly 1 head.
    - ii 2 heads.
    - iii No heads.
    - iv 1 head and 1 tail.
- 3 An archer makes two attempts to hit a target, and the probability that he hits the target on any one attempt is  $\frac{1}{3}$ . Find the probability that the archer will miss the target on both attempts.
- 4 A computer generates two random numbers between 1 and 100 (inclusive). Using the product rule, find the probability that the two numbers are:
  - a The same.
  - b Different.
  - c 89 and 83 in that order.
  - d 89 and 83 in any order.
  - e 14 and 36 in any order.
  - f Not 16 and 95.
- 5 A coin is tossed, a die is rolled, and a card is randomly selected from a normal deck of cards.
  - a Find the total number of outcomes.
  - b Find the probability that the coin lands on heads, the die lands on 6 and an Ace of Hearts is selected from the deck of cards.
  - c Find the probability that the coin lands on tails, the die lands on an even number, and the card selected is red.
- 6 In a game of Monopoly, rolling a double means rolling the same number on both dice. When you roll a double this allows you to have another turn. Find the probability that Sarah rolls:
  - a A double 1.
  - b A double 5.
  - c Any double.
  - d Two doubles in a row.

7 Christa enters a competition in which she guesses the 3-digit code (from 000 to 999) which cracks open a vault containing one million dollars. If the 3-digit number to open the vault is randomly generated by a computer, find the probability that it is:



- a An odd number.
- b An even number (including 000).
- c A number greater than 123.
- d A number divisible by 10.
- e A number less than 321.

8 A fair die is rolled twice. Find the probability of:

- a Rolling a 6 and a 1 in any order.
- b Rolling a 6 and then a 1.

9 A pile of playing cards has 4 diamonds and 3 hearts. A second pile has 2 diamonds and 5 hearts. One card is selected at random from the first pile, then the second.

- a Construct a probability tree of this situation with the correct probability on each branch.
- b Find the probability of selecting two hearts.

10 Students were asked what they are allergic to. The table shows the results:

|                       | Allergic to nuts | Not allergic to nuts |
|-----------------------|------------------|----------------------|
| Allergic to dairy     | 14               | 12                   |
| Not allergic to dairy | 25               | 25                   |

If a student is chosen at random, find the probability that:

- a The student is allergic to dairy.
- b The student has an allergy.
- c The student does not have an allergy.
- d The student is also allergic to dairy if he or she is allergic to nuts.



## Dependent events

- 11** Mae deals two cards from a normal deck of cards. Calculate the probability that she deals:
- a** Two 10s
  - b** Two red cards
  - c** Two diamonds
  - d** A 10 of spades and an Ace of diamonds, in that order
  - e** A 10 of clubs and an Ace of spades, in any order
- 12** A carton contains a dozen eggs, of which 4 contain no yolk. If 3 eggs are chosen at random for a cake, find the probability that they all have no yolk.
- 13** From a set of 10 cards numbered 1 to 10, two cards are drawn at random without replacement. Find the probability that:
- a** Both numbers are even.
  - b** One is even and one is odd.
  - c** The sum of the two numbers is 12.

## With or without replacement

- 14** Eileen randomly selects two cards, with replacement, from a normal deck of cards. Find the probability that:
- a** The first card is a queen of spades and the second card is a 4 of clubs.
  - b** The first card is spades and the second card is a 4.
  - c** The first card is a Queen and the second card is black.
  - d** The first card is not a 7 and the second card is not Clubs.
- 15** Beth randomly selects three cards, with replacement, from a normal deck of cards. Find the probability that:
- a** The cards are queen of diamonds, king of spades, and king of diamonds, in that order.
  - b** The cards are all black.
  - c** The first card is a 9, the second card is a heart and the third card is red.
  - d** The cards are all hearts.



- 16** A chess player is placed into a draw where in each match he has a 30% chance of winning. Find the probability that:
- a** He wins his first two matches.
  - b** He wins his first three matches.
  - c** He wins his first two matches and then loses his third match.
- 17** Amelia randomly selects two cards, with replacement, from a normal deck of cards. Find the probability that:
- a** Both cards are red
  - b** Both cards are the same color.
  - c** Both cards are of different colors.
- 18** A bag contains the letters A, B, C, D and E. Two letters are randomly drawn out of the bag without replacement.
- a** Construct a probability tree showing all possible outcomes.
  - b** Find the probability of the following:
    - i** A is drawn followed by B.
    - ii** Two A's are drawn.
    - iii** C is not drawn.
    - iv** D is drawn.
    - v** E is drawn and B is not.
    - vi** C or D is drawn.
    - vii** Neither C nor D are drawn.